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SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and are accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 8

Derivatives of Inverse Hyperbolic Functions

$$\text{Let } y = \sinh^{-1} x$$

$$\sinh y = x \quad \cosh y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = \frac{1}{\cosh y}$$

$$\text{Let } y = \cosh^{-1} x$$

$$\cosh y = x \quad \sinh y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = \frac{1}{\sinh y}$$

$$= \frac{1}{\sqrt{1 - \cosh^2 x}} = \frac{1}{\sqrt{1 - x^2}}$$

$$\text{Let } y = \tanh^{-1} x$$

$$\tanh y = x \quad \sec^2 y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = \frac{1}{\sec^2 y}$$

$$= \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}$$

$$\text{Let } y = \coth^{-1} x$$

$$\coth y = x \quad -\csc^2 y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = \frac{1}{\csc^2 y}$$

$$= \frac{1}{\coth^2 y - 1} = \frac{1}{x^2 - 1}$$



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$$\text{Let } y = \sec h^{-1} x$$

$$\sec hy = x \quad - \sec hy \tanh y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = -\frac{1}{\sec hy \tanh y}$$

$$\text{Let } y = \csc h^{-1} x$$

$$\csc hy = x \quad - \csc hy \coth y \frac{dy}{dx} = 1, \quad \frac{dy}{dx} = \frac{1}{-\csc hy \coth y}$$

$$\frac{dy}{dx} = -\frac{1}{\csc hy \coth y} = \frac{1}{|x|\sqrt{1+x^2}}$$

Example $y = \sinh^{-1} 2x$

Solution: $\sinh y = 2x \quad \cosh y \frac{dy}{dx} = 2, \quad \frac{dy}{dx} = \frac{2}{\cosh y}$

$$\frac{2}{\sqrt{1+\sinh^2 x}} = \frac{2}{\sqrt{1+4x^2}}$$

□

Example $y = \coth^{-1} \sqrt{x^2 + 1}$

Solution: $\coth y = (x^2 + 1)^{1/2} \quad -\csc h^2 y \frac{dy}{dx} = 1/2 (2x) (x^2 + 1)^{-1/2},$

$$\frac{dy}{dx} = \frac{-x}{-\csc h^2 y (x^2 + 1)^{1/2}} = \frac{-x}{(\coth^2 y - 1)(x^2 + 1)^{1/2}}$$



$$\frac{dy}{dx} = \frac{-x}{(x^2+1-x)(x^2+1)^{1/2}} = \frac{-1}{x(x^2+1)^{1/2}}$$

□

EXERCISE 1. ✎ $y = \ln(\tanh^{-1} x)$

Exercise

Find the derivatives of

- (1) $f(x) = \tanh^{-1} \sqrt{x}$
- (2) $f(x) = \cosh^{-1}(x^2 + 1)$
- (3) $f(x) = \csc h^{-1} e^x$
- (4) $f(x) = \sinh^{-1}(\ln x)$



Solutions to Exercises

Exercise 1. $\frac{dy}{dx} = \frac{1}{(1-x^2) \tanh^{-1} x}$

Exercise 1